Lecture 8

January 20, 2026

These notes should be referred to in continuation of Lecture 7 notes. Equation numbers are also in continuation of Lecture 7.

Important Remarks

- If we write $h(X) = |X|$, we see that $E(X)$ exists if and only if $E|X|$ does.
- A random variable X is symmetric about a point 'a' if

$$P(X \geq a + x) = P(X \leq a - x) \; \forall x.$$

In terms of CDF $F(x)$, this means that if

$$F(a - x) = 1 - F(a + x) + P(X = a + x)$$

holds for all $x \in \mathbb{R}$, we say that the CDF $F(x)$ is symmetric with 'a' as the center of symmetry.

Important Remarks (Contd.)

- If $a = 0$, then for every x,

$$F(-x) = 1 - F(x) + P(X = x).$$

- In particular, if X is a continuous random variable, X is symmetric with center 'a' if and only if the PDF $f(x)$ satisfies

$$f(a - x) = f(a + x)\ \forall x.$$

- If $a = 0$, we simply say that X is symmetric.

Important Remarks (Contd.)

- If 'a' and 'b' are constants and X is a random variable with $E|X| < \infty$, then $E|aX + b| < \infty$ and

$$E(aX + b) = aE(X) + b.$$

- In particular, $E(X - \mu) = 0$, where μ comes from (2) or (3).
- If X is bounded, i.e., if $P(|X| < M) = 1$, $0 < M < \infty$, then $E(X)$ exists.
- If $\{X \geq 0\} = 1$ and $E(X)$ exists, then $E(X) \geq 0$.

Important Result

Theorem 1

Let X be a random variable and g be a Borel-measurable function on $\mathbb{R}$. Let $Y = g(X)$. If X is discrete, then

$$E(Y) = \sum_{i=1}^{\infty} g(x_i) P(X = x_i) \tag{7}$$

in the sense that if either side of (7) exists, so does the other, and then the two are equal.
If X is continuous with PDF $f(x)$, then

$$E(Y) = \int g(x) f(x)\, dx \tag{8}$$

in the sense that if either of the two integrals converges absolutely, so does the other, and the two are equal.

Important Result (Contd.)

Proof of Theorem 1. In the discrete case, suppose that $P(X \in A) = 1$. If $y = g(x)$ is a one-to-one mapping of A onto some set B, then

$$P(Y = y) = P(X = g^{-1}(y)),\ y \in B.$$

We have

$$\sum_{x \in A} g(x)P(X = x) = \sum_{y \in B} yP(Y = y).$$

In the continuous case,

$$\int g(x)f(x)dx = \int_{\alpha}^{\beta} yf[g^{-1}(y)]\frac{d}{dy}g^{-1}(y)dy$$

by changing the variable to $y = g(x)$, where $\alpha < y < \beta$.
Thus

$$\int g(x)f(x)dx = \int_{\alpha}^{\beta} yh(y)dy.$$

Important Result (Contd.)

- The functions $h(x) = x^n$, where n is a positive integer, and $h(x) = |x|^\alpha$, where α is a positive real number, are of special importance.
- If $E(X^n)$ exists for some positive integer n, we call $E(X^n)$ the nth moment of X about the origin.
- If $E(|X|^\alpha) < \infty$ for some positive real number α, we call $E(|X|^\alpha)$ the αth absolute moment of X.
- We shall use the notation $\mu_n' = E(X^n)$.

Example 3

Let X is a discrete random variable with PMF

$$P(X = x) = \frac{1}{N}, \; x = 1, 2, \ldots, N.$$

Then

$$E(X) = \sum_{x=1}^{N} x \cdot \frac{1}{N} = \frac{N+1}{2},$$

$$E(X^2) = \sum_{x=1}^{N} x^2 \cdot \frac{1}{N} = \frac{(N+1)(2N+1)}{6}.$$

One can observe that, moments of all order exists here.

Example 4

Let X is a continuous random variable with PDF

$$f(x) = \begin{cases} \frac{2}{x^3}, & x \geq 1, \\ 0, & x < 1. \end{cases}$$

Then

$$E(X) = \int_1^\infty \frac{2}{x^2} dx = 2.$$

but

$$E(X^2) = \int_1^\infty \frac{2}{x} dx$$

does not exist.

Important Results

- There exist random variables for which all moments of a specified order exist but no higher-order moments do.
- If the moment of order t exists for a random variable X, moments of order $0 < s < t$ exist.
- If $h_1, h_2, \ldots, h_n$ are Borel-measurable functions of a random variable X and $E(h_i(X))$ exists for $i = 1, 2, \ldots, n$, then

$$E\left[\sum_{i=1}^{n} h_i(X)\right] = \sum_{i=1}^{n} E(h_i(X)).$$

Central Moments

- Let k be a positive integer and c be a constant. If $E(X-c)^k$ exists, we call it the moment of order k about the point c.
- If we take $c = E(X) = \mu$, which exists since $E|X| < \infty$, we call $E(X-\mu)^k$ the central moment of order k or the moment of order k about the mean. We shall write
$$\mu_k = E(X-\mu)^k.$$
- If we know $\mu_1', \mu_2', \ldots, \mu_k'$, we can compute $\mu_1, \mu_2, \ldots, \mu_k$ and conversely.

Relation between μ_k and μ'_k

$$\mu_k = \mu'_k - \binom{k}{1}\mu\mu'_{k-1} + \binom{k}{2}\mu^2\mu'_{k-2} - \cdots + (-1)^k\mu^k.$$

$$\mu'_k = \mu_k + \binom{k}{1}\mu\mu_{k-1} + \binom{k}{2}\mu^2\mu_{k-2} + \cdots + \mu^k.$$

Variance

- If $E(X^2)$ exists, we call $E(X-\mu)^2$ the variance of X, and we write

$$\sigma^2 = \mathrm{Var}(X) = E(X-\mu)^2.$$

- The quantity σ is called the standard deviation of X.
- We can also write

$$\sigma^2 = \mu_2 = E(X^2) - (E(X))^2.$$

- $\mathrm{Var}(X) = 0$ if and only if X is degenerate.
- $\mathrm{Var}(X) < E(X-c)^2$ for any $c \neq E(X)$.
- $\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$.

Standardized Random Variable

Let $E|X|^2 < \infty$. Then we define

$$Z = \frac{X - E(X)}{\sqrt{\mathrm{Var}(X)}} = \frac{X - \mu}{\sigma}.$$

One can observe that $E(Z) = 0$ and $\mathrm{Var}(Z) = 1$.
We call Z a standardized random variable.

Quantile

A number q satisfying

$$P(X \leq q) \geq p, \quad P(X \geq q) \geq 1 - p,\ 0 < p < 1,$$

is called a quantile of order p [or $(100p)$th percentile] for the random variable X.
If q is a quantile of order p for a random variable X with CDF F, then

$$p \leq F(x) \leq p + P(X = q).$$

If $P(X = q) = 0$ (in particular if X is continuous), a quantile of order p is a solution of the equation $F(x) = p$.

Median

Let X be a random variable with CDF F. A number m satisfying

$$\frac{1}{2} \leq F(x) \leq \frac{1}{2} + P(X = m)$$

or equivalently

$$P(X \leq m) \geq \frac{1}{2} \quad \text{and} \quad P(X \geq m) \geq \frac{1}{2}$$

is called a median of X.
If F is a symmetric CDF, the centre of symmetry is clearly the median of the CDF F.

Example 5

Let X be a random variable with PMF

$$P(X=-2)=P(X=0)=\frac{1}{4}, \quad P(X=1)=\frac{1}{3}, \quad P(X=2)=\frac{1}{6}.$$

Then

$$P(X \leq 0)=\frac{1}{2} \quad \text{and} \quad P(X \geq 0)=\frac{3}{4}>\frac{1}{2}.$$

In fact, if m is any number such that $0<m<1$, then

$$P(X \leq m)=\frac{1}{2}, \quad P(X \geq m)=\frac{1}{2},$$

and it follows that every m, $(0 \leq m<1)$, is a median of X.

Example 5 (Contd.)

If $p = 0.2$, the quantile of order p is $q = -2$, since

$$P(X \leq -2) = \frac{1}{4} > p \quad \text{and} \quad P(X \geq -2) = 1 > 1 - p.$$